

3. **RJ45** – short for **Registered Jack-45**. RJ-45 is an 8-pin connection used for Ethernet network adapters. RJ-45 resembles the RJ-11 or 6-pin connector used with telephones in United States (they are completely different). This picture is an RJ-45 connector separated from the cable.



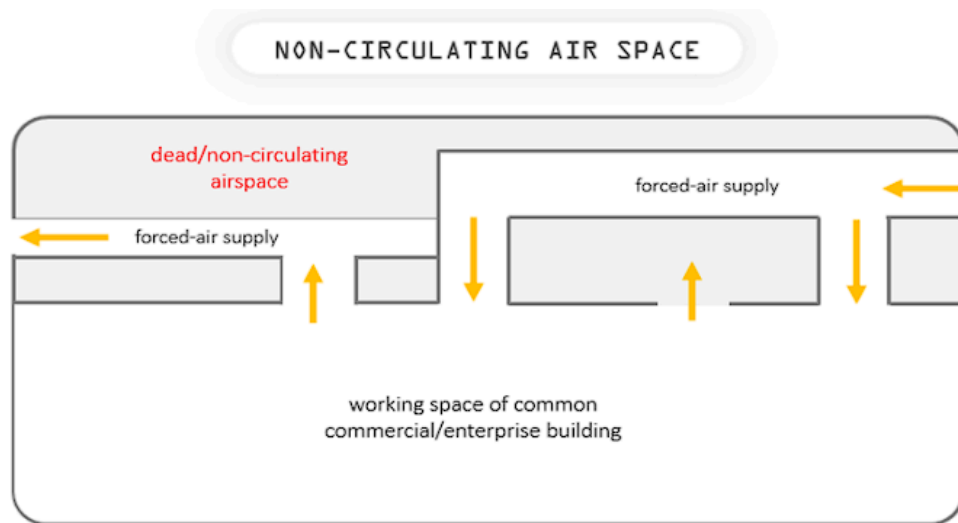
Figure 3. Registered Jack-45

4. **Network cable tester** – is a device used to test the strength and connectivity of a particular type of other wired assemblies. There are different types of cable testers, each able to test specific type of cable or wire. A cable tester can test whether a cable or wire is set up properly, connected correctly, and the communication strength between the source and destination.

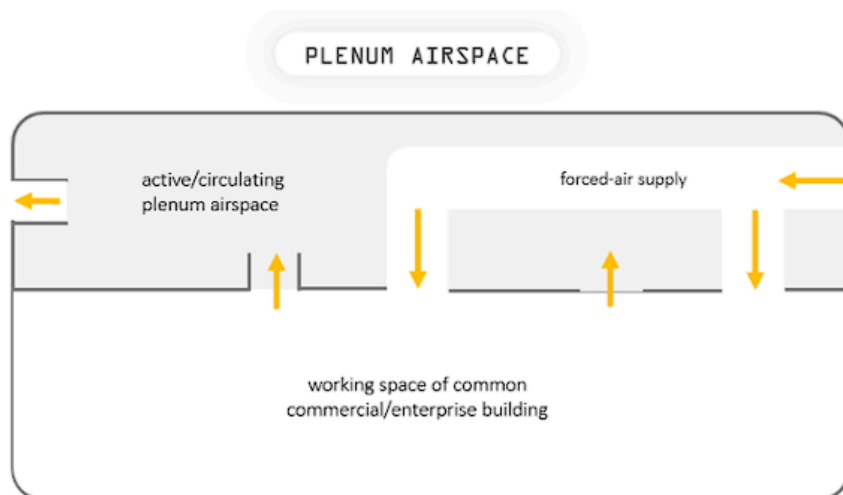
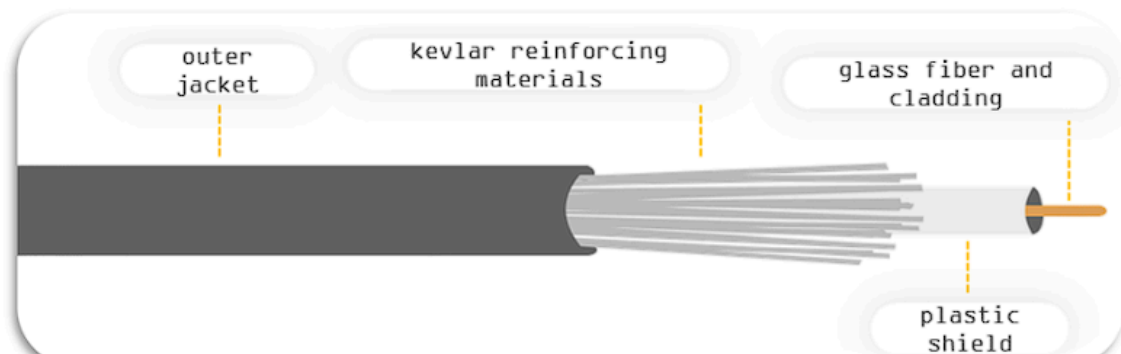


Figure 4. Network cable tester

5. **UTP cable** – short for unshielded twisted pair cable. It is the most popular type of network cable which consists of two (2) unshielded wires twisted around each other. Due to its low cost, UTP cabling is widely used for local-area networks (LANs) and telephone communications.



4. **Fiber Optic Cable** – is a cable that contains optical fibers (*usually glass*) coated in plastic that are used to send data by pulses of light.



Types of Fiber Optic Cable

- **Single-mode Fiber Cable** – allows only one mode (or wavelength) of light to propagate through the fiber. It is capable of higher bandwidth and greater distances than multimode, and it is often used for campus backbones.
- **Multi-mode Fiber Cable** – allows multiple modes of light to propagate through the fiber. It is often used for workgroup applications and intrabuilding applications such as risers. It uses light-emitting diodes (LEDs) as a light-generating device. The maximum cable length is 2 kilometers.

	Types of Fiber Optic Cable Differences	
	Single-mode Fiber Cable	Multi-mode Fiber Cable
Length	Long distance applications	Short distance applications
Diameter	Smaller	Bigger
Light Source	Light Emitting Diode	Laser Diode
Bandwidth	Limited	Unlimited
Color	Yellow	Orange (sometimes Cyan)

COMMON NETWORK CABLE

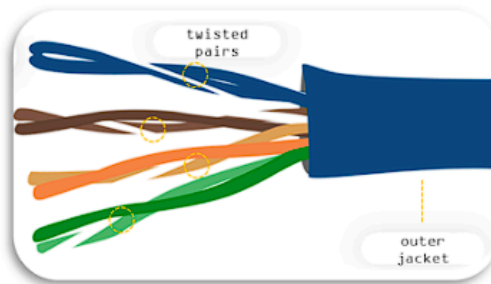
BACKGROUND INFORMATION FOR LEARNERS

Network Cable (or Network Media) is the actual path over which an electrical signal travels as it moves from one component to another. This Learning Activity Sheet will describe the common types of network media, including twisted-pair cable, coaxial cable, fiber-optic cable, and wireless.

These cables are used to transfer and connect information or data between two or more **computers, routers, switches, and storage area networks.**

Types of Network Media:

1. **Twisted-pair Cable** – is a type of cabling that is used for telephone communications and most modern Ethernet networks. There are two (2) types of twisted-pair cable which are:
 - a. **Unshielded Twisted Pair** – or simply UTP, is a medium that is composed of pairs of wires. UTP cable is used in a variety of networks. Each of the eight (8) individual copper wires in UTP cable is covered by an insulating material.



- b. **Shielded Twisted Pair** – it is a cable with the combination techniques of shielding, cancellation, and wire-twisting. Each pair of wires is wrapped in a metallic foil. The four pairs of wires then are wrapped in an overall metallic braid or foil.

Remember, the color-coding must be done from BOTH ENDS of the cable. Thus, you will need two (2) pieces of RJ-45 in order for you to make a straight-through cable.

TOOLS AND MATERIALS NEEDED:

1. **Crimping tool** – is a device used to conjoin two (2) pieces of metal by deforming one or both of them in a way that causes them to hold each other. The result of the tool's work is called crimp. In configuring cable, crimping tool is used to lock/join RJ-45 and UTP cable.

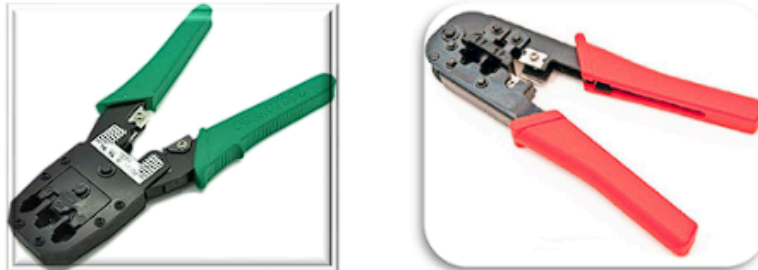


Figure 1. Crimping tool

2. **Wire stripper** – is a tool designed to remove the protective covering (jacket) off a cable to expose the inner wires. Since inner wires comes in different shapes, there are dozens of wire strippers available. In the picture, is an example of a wire stripper that strips cables when placed in-between the blade and the



holder and spun around the cable until the cable jacket can be pulled off.

Figure 2. Wire stripper

Figure 3. T568 B color-coding configuration

Figure 3 identifies on how you will pair the wires by using this color-coding configuration. The color-coding configuration is as follows:

White Orange	Paired with	Orange
White Green	Paired with	Blue
White Blue	Paired with	Green

White Brown	Paired with	Brown
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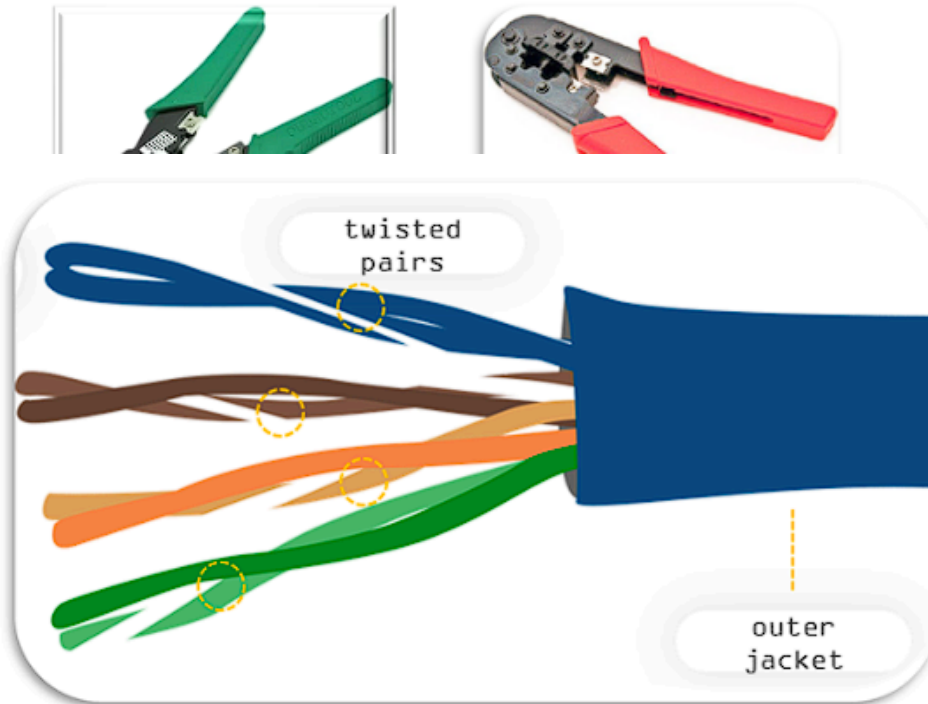


Figure 5. UTP cable

PROCEDURES:

1. Strip the cable jacket back one full inch.
2. Untwist the wires back to within 1/8 inch of the jacket.
3. Arrange the wires in the order in which you want to crimp them. (T568 B each end)
4. Grasp the wires firmly, between your thumb and forefinger, flatten them, and even wiggle them a bit to take out the curliness.
5. Concentrate your efforts on the bottom 1/2 inch. The wires must lay flat and together, aligned as close as possible.
6. While holding the wires firmly, cut off the wires 1/2 inch from the cables jacket. Cut the wires with some sharp wire strippers or even crimping tool. Avoid wire cutters that flatten the ends of the wires insulating material. (this makes inserting the wires very difficult)
7. Insert the wires into RJ-45. Make sure that the wires stay LINED-UP. The wires should reach the end of the little tube they are in.
8. The covering or jacket should even with the end of the first indent (Small bump line on the lower part of RJ-45). This must be done because it's a strain relief for the cable.
9. Insert it into the crimping tool, and CRIMP IT. You cannot un-crimp once you crimp RJ-45 to UTP cable.
10. Use the network cable tester to test the Ethernet cable. If all the lights blink, it's a good network cable. But if one or more lights do not blink, cut the RJ-45 out because once you crimp it, you cannot use that RJ-45 anymore.

CONFIGURING STRAIGHT-THROUGH ETHERNET CABLE

BACKGROUND INFORMATION FOR LEARNERS

STRAIGHT THROUGH CONFIGURATION

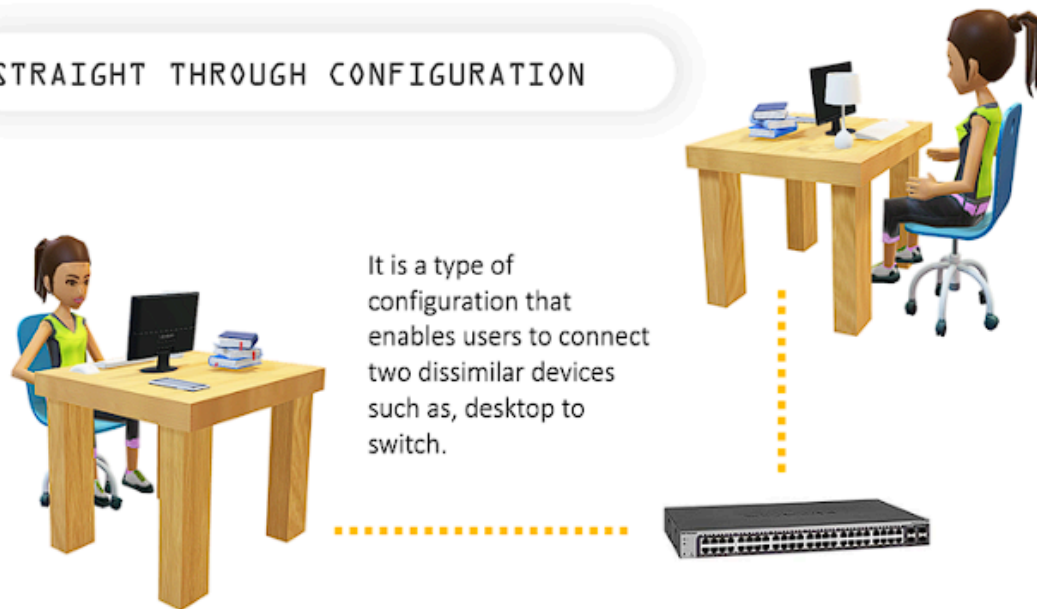
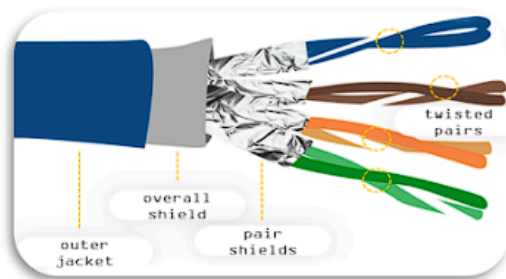
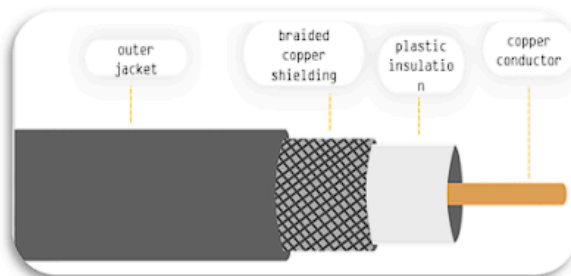


Figure 1. Example of a Straight-through configuration.

Analyze the image above. Are the two (2) computers connected? How? The computers were connected via Ethernet cable into a switch. This is one example of straight-through cable configuration because computer and switch are different devices and only a straight-through cable can connect them.



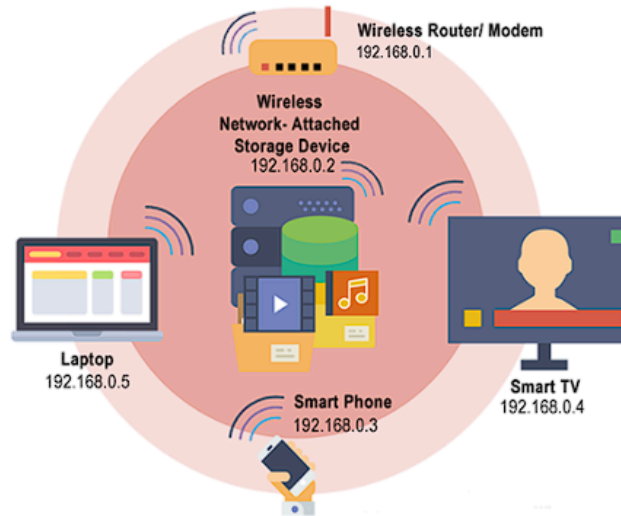
2. **Coaxial Cable** – it is a cable used in the transmission of video, communications, and audio. However, these cables are also used in networks and what allow a broadband cable Internet connection using in a cable modem. (e.g. Sky Movie Cable Internet)



3. **Plenum Cable** – is the cable that runs in plenum spaces in building. In building construction, a plenum is a separate space provided for air circulation for heating, ventilation, air-conditioning (sometimes referred to as HVAC), typically in the space between the structural ceiling and a drop-down ceiling.

	Types of Fiber Optic Cable Differences	
	Single-mode Fiber Cable	Multi-mode Fiber Cable
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Light Source	Light Emitting Diode	Laser Diode
Bandwidth	Limited	Unlimited
Color	Yellow	Orange (sometimes Cyan)
Price	Expensive	Cheap

5. **Wireless Communication** – it uses radio frequencies (RF) or infrared (IR) waves to transmit data between devices on a LAN. For wireless LANs, a key component is the wireless hub, or access point, used for signal distribution.



CONFIGURING CROSSOVER ETHERNET CABLE

BACKGROUND INFORMATION FOR LEARNERS

Last week, we learn what is a straight-through cable configuration and how to make it. We also learn that the purpose of straight-through cable is to connect “different” devices like computer to router junction. In this Learning Activity Sheet, we will be going to tackle the Crossover configuration cable.

CROSSOVER CONFIGURATION

It is a type of configuration that enables users to connect similar devices such as, desktop to another desktop.



Figure 1. Example of a Crossover configuration.

Table 1. T568 A Color-Coding Configuration

Other End of the Cable		
White Orange	Paired with	Orange
White Green	Paired with	Blue
White Blue	Paired with	Green
White Brown	Paired with	Brown

Table 2. T568 B Color-Coding Configuration

Remember, the color-coding must be done **differently on the two ends** of the cable. The one end can be T568 A and the other end must be T568 B.

The **Tools and Procedures** are the same as how straight-through cable was made. The only difference was the color-coding procedure.

IMPORTANT NOTES:

- A crossover cable has **DIFFERENT ENDS**.
- Crossover is used to connect two (2) similar Ethernet devices without hub, switch, or router. (e.g. laptop to laptop)